

CS105 Discrete Structures: Induction

Pop Quiz 1 Solutions

1.1

Solution

Base case: For $n = 0$, we have an equilateral triangle with $4^0 \cdot 3 = 3$ sides, which is correct.

1 mark.

Induction hypothesis: Assume the statement is true for some $n = k$, i.e., the number of sides is $4^k \cdot 3$.

1 mark.

Induction step: We need to show it holds for $n = k + 1$, i.e., the number of sides is $4^{k+1} \cdot 3$. When we transform K_k into K_{k+1} , each side of K_k is replaced by 4 sides of K_{k+1} , hence the number of sides becomes $4 \cdot (4^k \cdot 3) = 4^{k+1} \cdot 3$. This completes the induction step.

1 mark.

Hence by the principle of mathematical induction, the statement is true for all $n \geq 0$.

1.2

Solution

The negation is $\exists n \in \mathbb{N}$ such that number of sides of K_n is not equal to $4^n \cdot 3$.

0.5 mark.

Negation is false as we proved by induction above that the number of sides of K_n is always $4^n \cdot 3$ for all $n \geq 0$.

0.5 mark.

2.1

Solution

Soln 1 mark, Proof 1 mark. Using induction, we will prove that the perimeter P_n is $3 \cdot \left(\frac{4}{3}\right)^n$.

Base case: for $n = 0$ we have our equilateral triangle with 3 sides of length 1, hence it has perimeter 3. The formula $3 \cdot \left(\frac{4}{3}\right)^n$ at $n = 0$ also evaluates to 3, hence it is correct.

Induction hypothesis: Let P_i equal $3 \cdot \left(\frac{4}{3}\right)^i$ for some i . We will show that P_{i+1} is equal to $3 \cdot \left(\frac{4}{3}\right)^{i+1}$.

Induction step: In transforming K_n to K_{n+1} , we replace each side with 4 sides of length $\frac{1}{3}$ rd the original side. If the original length of the side was s , it is replaced by sides of total length $\frac{4s}{3}$. Since each side is replaced in this way, the total perimeter is also multiplied by $\frac{4}{3}$. Since according to our induction hypothesis the initial perimeter is $3 \cdot \left(\frac{4}{3}\right)^i$, it will now be $3 \cdot \left(\frac{4}{3}\right)^i \cdot \frac{4}{3} = 3 \cdot \left(\frac{4}{3}\right)^{i+1}$. This is what we needed to prove P_{i+1} to equal to.

Hence proved by induction P_n equals $3 \cdot \left(\frac{4}{3}\right)^n$.

2.2

Solution

Soln 1 mark, proof 1 mark Using induction, we will prove that the area A_n is $\frac{\sqrt{3}}{4} \left(\frac{8}{5} - \frac{3}{5} \left(\frac{4}{9}\right)^n\right)$.

Base case: For $n = 0$, we have an equilateral triangle of side length 1, whose area is $\frac{\sqrt{3}}{4}$. Inputting $n = 0$ into the formula, we get $A_0 = \frac{\sqrt{3}}{4} \left(\frac{8}{5} - \frac{3}{5} \left(\frac{4}{9}\right)^0\right) = \frac{\sqrt{3}}{4} \left(\frac{8}{5} - \frac{3}{5}\right) = \frac{\sqrt{3}}{4} \left(\frac{5}{5}\right) = \frac{\sqrt{3}}{4}$. Hence it is correct.

Induction hypothesis: Let $A_i = \frac{\sqrt{3}}{4} \left(\frac{8}{5} - \frac{3}{5} \left(\frac{4}{9}\right)^i\right)$ for some i . We will show that $A_{i+1} = \frac{\sqrt{3}}{4} \left(\frac{8}{5} - \frac{3}{5} \left(\frac{4}{9}\right)^{i+1}\right)$.

Induction step: In transforming K_i to K_{i+1} , we increase the area by adding an equilateral triangle for every line segment. The side length of the added triangle is $\frac{1}{3}$ of the segment length. Since the total perimeter at step i is $P_i = 3 \cdot \left(\frac{4}{3}\right)^i$ and the number of sides is $3 \cdot 4^i$, the length of each side is $\left(\frac{1}{3}\right)^i$. Hence, the side length of the equilateral triangle added is $\frac{1}{3} \cdot \left(\frac{1}{3}\right)^i = \left(\frac{1}{3}\right)^{i+1}$. This makes the area of one small triangle equal to $\frac{\sqrt{3}}{4} \left(\frac{1}{3}\right)^{2(i+1)} = \frac{\sqrt{3}}{4} \cdot \frac{1}{9^{i+1}}$. Since this area is added for each of the $3 \cdot 4^i$ line segments, we multiply to get the total added area:

$$\text{Area Added} = (3 \cdot 4^i) \cdot \frac{\sqrt{3}}{4 \cdot 9^{i+1}} = \frac{3\sqrt{3}}{4} \cdot \frac{4^i}{9 \cdot 9^i} = \frac{\sqrt{3}}{12} \left(\frac{4}{9}\right)^i$$

The new total area is $A_{i+1} = A_i + \text{Area Added}$. Inputting our formula for A_i :

$$A_{i+1} = \left[\frac{\sqrt{3}}{4} \left(\frac{8}{5} - \frac{3}{5} \left(\frac{4}{9}\right)^i\right) \right] + \frac{\sqrt{3}}{12} \left(\frac{4}{9}\right)^i$$

We can factor $\frac{\sqrt{3}}{4}$ from the second term, since $\frac{\sqrt{3}}{12} = \frac{\sqrt{3}}{4} \cdot \frac{1}{3}$:

$$A_{i+1} = \frac{\sqrt{3}}{4} \left[\left(\frac{8}{5} - \frac{3}{5} \left(\frac{4}{9}\right)^i\right) + \frac{1}{3} \left(\frac{4}{9}\right)^i \right]$$

Now, combine the terms containing $\left(\frac{4}{9}\right)^i$:

$$\begin{aligned} A_{i+1} &= \frac{\sqrt{3}}{4} \left[\frac{8}{5} + \left(\frac{1}{3} - \frac{3}{5}\right) \left(\frac{4}{9}\right)^i \right] \\ A_{i+1} &= \frac{\sqrt{3}}{4} \left[\frac{8}{5} - \frac{4}{15} \left(\frac{4}{9}\right)^i \right] = \frac{\sqrt{3}}{4} \left[\frac{8}{5} - \frac{9}{15} \left(\frac{4}{9}\right) \left(\frac{4}{9}\right)^i \right] \end{aligned}$$

This gives the required formula:

$$A_{i+1} = \frac{\sqrt{3}}{4} \left(\frac{8}{5} - \frac{3}{5} \left(\frac{4}{9}\right)^{i+1}\right)$$

Hence proved by induction A_n equals $\frac{\sqrt{3}}{4} \left(\frac{8}{5} - \frac{3}{5} \left(\frac{4}{9}\right)^n\right)$